



higher profits through more efficient design processes. BIG-Net uses deep learning to analyse SVG images of products, accurately detect brand elements, and successfully identify brand consistencies in product images. It distinguished between cellphone brands like Apple and Samsung by identifying specific features with 100% accuracy. BIGNet also accurately differentiates between ten car brands, recognising luxury vehicles through branding. This method reduces the need for domain experts and marks a significant shift from manually identifying brand features. Researchers see potential for BIGNet in 3D imaging and beyond, highlighting its impact on product design and brand management by offering unparalleled efficiency and precision.

Vanadium flow batteries could revolutionise large scale energy storage

Scientists at the Dalian Institute of Chemical Physics, Chinese Academy of Sciences, have developed a high-



70kW-level vanadium flow battery stack
(Source: <https://interestingengineering.com>)

power vanadium flow battery stack, potentially revolutionising renewable energy storage for large-scale applications. Vanadium flow batteries, notable for their longevity and safety, have faced limited adoption due to high costs. The team's

70kW stack doubles the power of previous 30kW stacks using new materials, achieving a volume power density of 130kW/m³ and reducing costs by 40%. It maintains over 80% energy efficiency across different power settings and shows minimal efficiency loss after 1200 cycles. This innovation could double the power of 6.1-metre (20-foot) energy storage units to 500kW without significant size or cost increases, paving the way for commercialising vanadium flow batteries and enhancing sustainable energy storage.

Helium 1 chip prototype showcases quantum supremacy with fewer resources

Researchers at the research institute Inria have introduced a quantum error correction architecture using low-density parity-check (LDPC) codes on cat qubits, advancing towards scalable, fault-tolerant quantum computing. This architecture optimises gate implementation and short-range connectivity on quantum chips, reducing the complexity of quantum error correction. It offers a tenfold increase in error correction efficiency, enabling the operation of 100



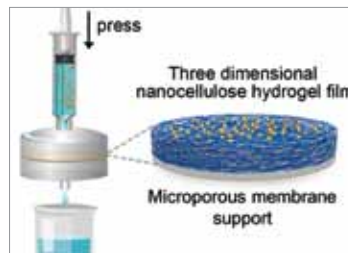
Cat Qubit Quantum error correction
(Source: <https://thequantuminsider.com>)

logical qubits with just 1500 physical cat qubits. Cat qubits, inspired by Schrödinger's Cat Paradox, effectively reduce bit-flip errors. This setup allows for executing advanced algorithms, such as Shor's algorithm, with fewer than 100,000

physical qubits—significantly less than the millions previously thought necessary. The Helium 1 chip prototype by Alice & Bob showcases the potential to achieve quantum supremacy with fewer resources, indicating a major leap towards exceeding classical computing power.

New water filtration method removes nearly all particles larger than 10nm

Researchers at the University of Texas have developed a new water filtration system, significantly enhancing clean drinking water access for over two billion people world-



Injectable water filtration system
(Source: <https://news.utexas.edu>)

wide, especially in remote and underdeveloped regions with particle-contaminated water. Published in Nature Sustainability, this portable system filters out nearly 100% of particles larger than 10 nanometres (nm), outperforming existing

methods that capture 40% to 80% of such particles. It uses a web of nano cellulose fibres, enabling easy water purification via syringe, delivering clean water instantly. Affordable, user-friendly, and sustainable, the system uses biodegradable hydrogel films for up to 30 uses. Effective against various types of contaminated water, including those with microplastics, the syringe-based prototype processes up to 1.5 litres of water. This marks a significant advancement towards achieving the United Nations' Sustainable Development Goals for global safe drinking water access.

Quantum computing poised to improve fibre network operations

Cinfo and Kipu Quantum are collaborating to enhance the resilience of Galicia's optical fibre network, operated by R, using quantum computing. This project identifies critical